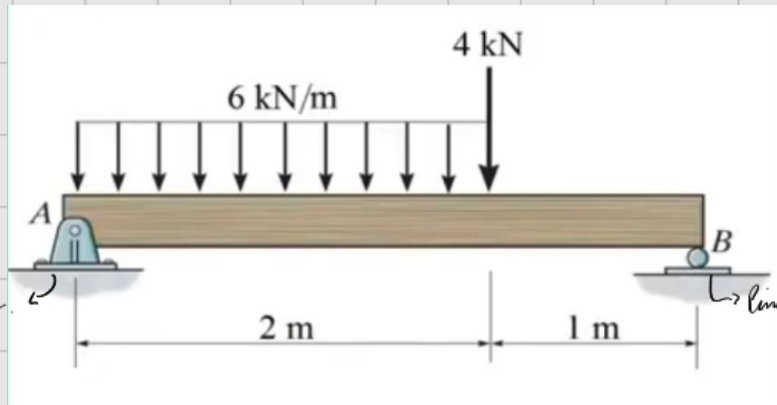


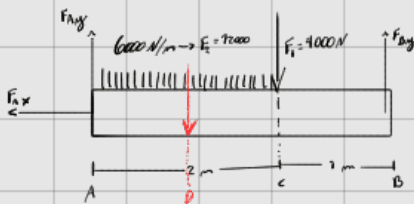
Hugo Antenor Santos



linha, resistindo x e y.

linha, resistindo apenas em y

* Diagrama de carga livre:



* Como o carga é retangular: $E_q = w \cdot l$
 $E_q = 6000 \cdot 2 = 12000$ $x(F_1) = \frac{x}{2} = \frac{2}{2} = 1m$ (Centro do carga)

$$\sum F_x = 0$$

$$\sum F_y \rightarrow F_{Ay} + F_{By} - 12000 - 4000 = 0$$

$$F_{Ay} = 16000 - F_{By}$$

$$\rightarrow F_{Ay} = 16000 - 6666,66 = 9333,33N$$

* Momentos de reação do carga:

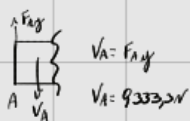
$$\text{PINO A: } \sum M_A = 0 \rightarrow F_{By} \cdot 3 - 4000 \cdot 2 - 12000 \cdot 1 = 0 \rightarrow 3F_{By} = 20000 \rightarrow F_{By} = 6666,66N$$

$$F_{By} = 6666,66N$$

$$F_{Ay} = 9333,33N$$

* Diagrama de força cortante

$$\sum F_y = 0$$

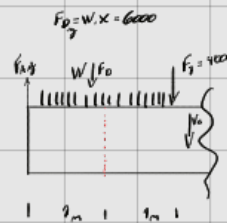


$$W = \frac{F}{x} \rightarrow F_D = W \cdot x = 6000x$$

$$-V_D - F_D + F_{Ay} = 0$$

$$V_D = 9333,33 - 6000x$$

$$V_D = 3333,33N$$



$$F_{Ay} - F_{By} - F_1 - V_0 = 0$$

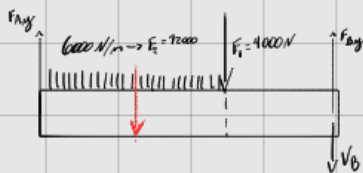
$$V_0 = 9333,33 - 6000 \cdot 2 - 4000$$

$$V_0 = -2667N$$

$$V_c = F_{Ay} - F_{By} - F_1$$

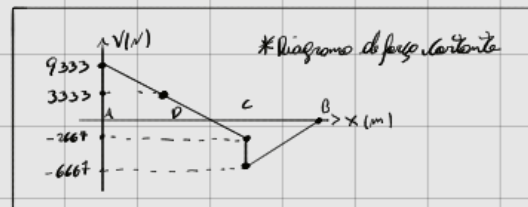
$$V_c = 9333,33 - 6000 \cdot 2 - 4000$$

$$V_c = -6667N$$



$$V_B = 9333 - 12000 - 4000 + 6666$$

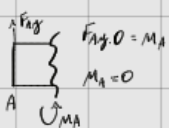
$$V_B = 0$$



* Diagrama de momento fletor: $\sum M_A = 0$

$$M = \frac{w \cdot x^2}{2} \rightarrow M = F \cdot d$$

$$M = (w \cdot x) \cdot \left(\frac{x}{2}\right)$$

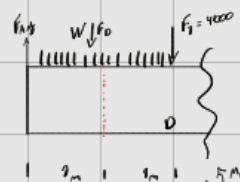


$$-F_{Ay} \cdot 0 + M_A = 0$$

$$M_A = 0$$

$$-F_{By} + \frac{w \cdot x^2}{2} + M_D = 0$$

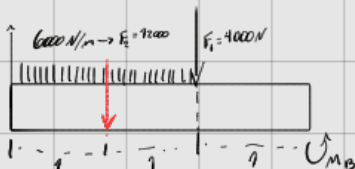
$$M_D = 6333,33N$$



$$M_0 + \frac{w \cdot x^2}{2} - A_y \cdot (x) = 0$$

$$M_0 = 9333,33 - \frac{6000 \cdot 4}{2}$$

$$M_0 = 6667N$$



$$M_B = 4000 \cdot 2 - 12000 \cdot 1 + 9333,33 = 0$$

$$M_B = 0$$

